# Multiple Linear Regression

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# Introduction

- Think about **multiple linear regression** (MLR) as an extension of **simple linear regression** (SLR)
  - SLR concepts largely stay the same but with more predictors and slightly different interpretations
- For this demo, I will use the `Hitters` data set stored in the `ISLR` package
  - Contains Major League Baseball (MLB) data from the 1986 and 1987 seasons

# Overview

- **Multiple Linear Regression** (MLR): Understanding the linear relationship between a response/dependent variable $Y$ using *multiple* $X$ variables
  - How do hours studied ($X_1$) and hours slept ($X_2$) linearly explain an exam grade ($Y$)?
  - How do square footage ($X_1$) and number of bedrooms ($X_2$) linearly explain a house price ($Y$)?

# Applying MLR to `Hitters` Model

- **SLR Model**: The linear relationship between a player's salary in thousands of USD ($Y$) in the 1987 season by their hits ($X$) in the 1986 season

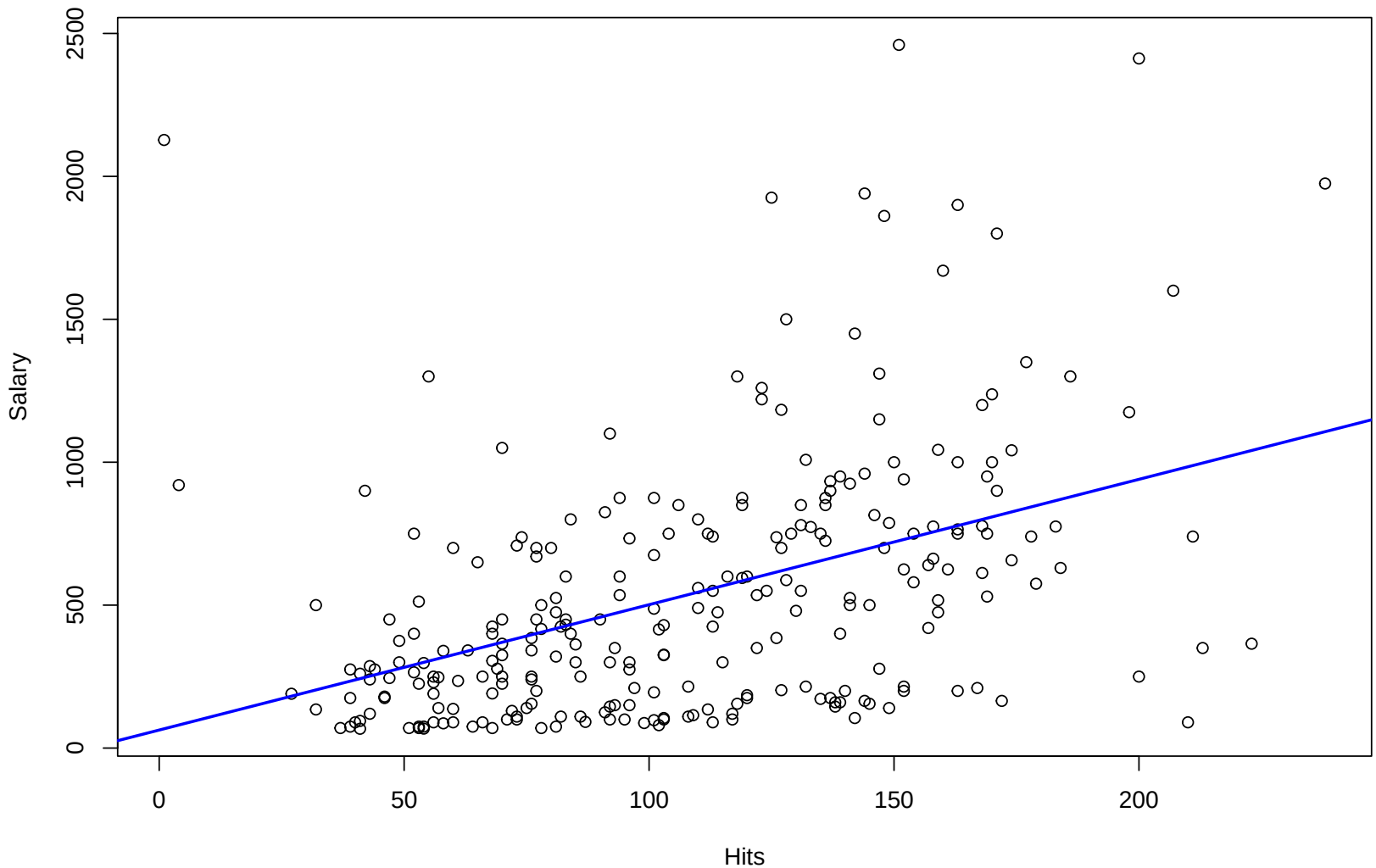
  $Y = \beta_0 + \beta_1 X + \epsilon$, where $\epsilon$ is the error in our prediction

- **MLR Model**: The linear relationship between a player's salary in thousands of USD ($Y$) in the 1987 season by their hits ($X_1$) in the 1986 season and years of MLB experience ($X_2$)

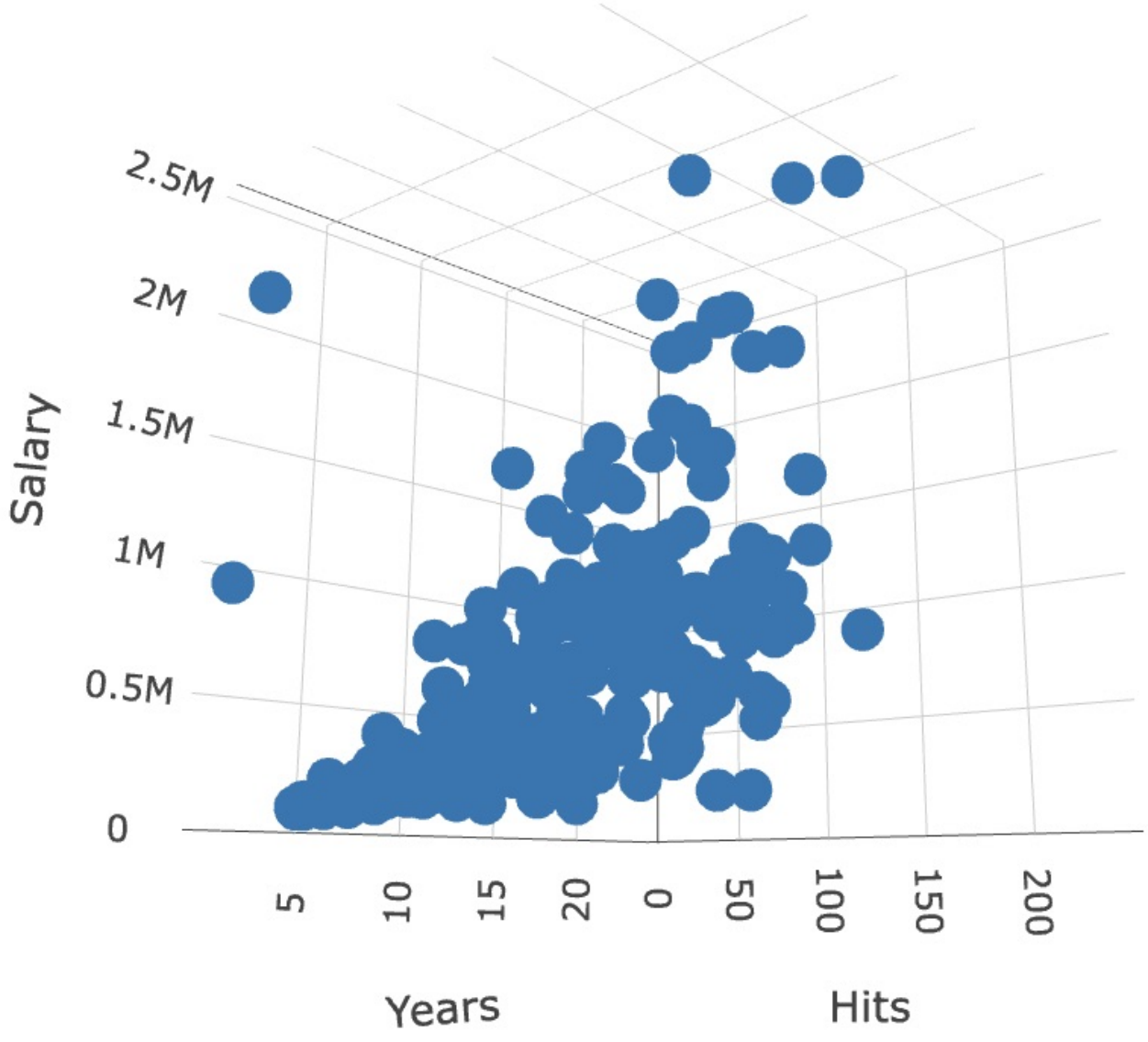
$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \epsilon$, where $\epsilon$ is the error in our prediction

# SLR Plot

```
plot(Salary ~ Hits, Hitters)
abline(lm(Salary ~ Hits, Hitters), col = "blue", lwd = 2)
```



# MLR Plot



# SLR Summary Table

```
summary(lm(Salary ~ Hits, Hitters))
```

```
Call:
lm(formula = Salary ~ Hits, data = Hitters)

Residuals:
    Min      1Q  Median      3Q     Max
-893.99 -245.63  -59.08  181.12 2059.90

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  63.0488    64.9822   0.970    0.333
Hits          4.3854     0.5561   7.886 8.53e-14 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 406.2 on 261 degrees of freedom
  (59 observations deleted due to missingness)
Multiple R-squared:  0.1924,    Adjusted R-squared:  0.1893
F-statistic: 62.19 on 1 and 261 DF,  p-value: 8.531e-14
```

- Note: (Intercept) ($\hat{\beta}_0$) does not have any stars next to it

# Coefficients

- **Estimate** ($\hat{\beta}$): The change in $Y$ given a one-unit change in $X$
- **Std. Error** ($SE$): The portion of the error in the model's predicted/fitted values ($\hat{Y}$) from their respective actual values ($Y$) attributed to $X$
- **t value** ($t$): The size of the estimate ($\hat{\beta}$) relative to the standard error ($SE$)
- **Pr(>|t|)** ($p$): The probability of observing values more extreme than in our data set ($t$)
  - Compared to a baseline alpha ($\alpha$) value, often 0.05
  - Used in **hypothesis testing** (next slide!)

# Hypothesis Testing

- **Hypothesis Test**: A test to determine whether we have *sufficient evidence* for an estimate ($\hat{\beta}$) being non-zero
  - The **null hypothesis** ($H_0$) states there is *no difference* between $\hat{\beta}$ and 0
    - $\hat{\beta} = 0$
  - The **alternative hypothesis** ($H_A$) states there is *a difference* between $\hat{\beta}$ and 0
    - $\hat{\beta} \neq 0$
- If $p \geq \alpha$, you would *fail to reject* $H_0$ and say $\hat{\beta}$ is *not significant*
- If $p < \alpha$, you would *reject* $H_0$ and say $\hat{\beta}$ is *significant*
- Significance is denoted by Signif. codes in the summary table

# R-Squared ($R^2$)

- The percentage of variance in $Y$ explained by the model
  - In our case, it is the percentage of variance in $Y$ explained by $X_1$ and $X_2$
- Shorthand name for the *coefficient of determination*
  - Calculated by squaring the *Pearson correlation coefficient* ($r$)
- $R^2$ can never decrease as you add more $X$ predictor variables
  - The *adjusted* $R^2$ adjusts accordingly, shrinking the *multiple* $R^2$ by the number of predictors

# F-statistic ($F$) & p-value ($p$)

- The F-statistic ($F$) tests whether at least one $X$ is significant in the model
  - 62.19 on 1 and 261 DF
    - $F = 62.19$
    - 1 *model* degree of freedom ($k$) is the number of $X$ predictors
    - 261 *residual* degrees of freedom ($n - (k + 1)$) is the difference in observed values $n$ and model parameters
- The p-value ($p$) follows the same setup as above but with a different hypothesis test
  - $H_0$: Every $\hat{\beta} = 0$
  - $H_A$: Any $\hat{\beta} \neq 0$
  - $p = 8.531\text{e}^{-14} < \alpha = 0.05$, so we *reject* $H_0$
    - At least one $\hat{\beta} \neq 0$

# MLR Summary Table

```
summary(lm(Salary ~ Hits + Years, Hitters))
```

```
Call:
lm(formula = Salary ~ Hits + Years, data = Hitters)

Residuals:
    Min      1Q  Median      3Q     Max
-838.06 -211.89  -43.40   76.07 2248.37

Coefficients:
             Estimate Std. Error t value Pr(>|t|)
(Intercept) -199.2510    67.4690  -2.953  0.00343 **
Hits           4.3124     0.5013   8.603 7.46e-16 ***
Years         36.9501     4.7187   7.831 1.24e-13 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 366.1 on 260 degrees of freedom
  (59 observations deleted due to missingness)
Multiple R-squared:  0.3465,    Adjusted R-squared:  0.3415
F-statistic: 68.94 on 2 and 260 DF,  p-value: < 2.2e-16
```

- Every predictor variable is significant!

# Interpreting Linear Relationships

- Intercept: $\hat{\beta}_0 =$ -199.25
  - We expect a player's 1987 salary in thousands of USD ($Y$) to be -199.25 with no years of experience and no hits during the 1986 season
- Hits: $\hat{\beta}_1 = 4.31$
  - We expect a player's 1987 salary in thousands of USD ($Y$) to increase by 4.31 for a one-unit increase in their number of hits during the 1986 season, holding years of experience constant.
- Years: $\hat{\beta}_2 = 36.95$
  - We expect a player's 1987 salary in thousands of USD ($Y$) to increase by 36.95 for a one-unit increase in years of experience, holding hits during the 1986 season constant.

# Confidence Intervals

- Let's look at the Hits predictor ($X_1$)
- To derive a confidence interval, use the `confint()` function in the `stats` package
- We are 95% confident that the true increase in player salary in the 1987 season in thousands of USD ($Y$) is between 3.33 and 5.3 for a one-unit increase in hits during the 1986 season.

```
confint(lm(Salary ~ Hits + Years, Hitters), "Hits", level = 0.95) # 95% CI
```

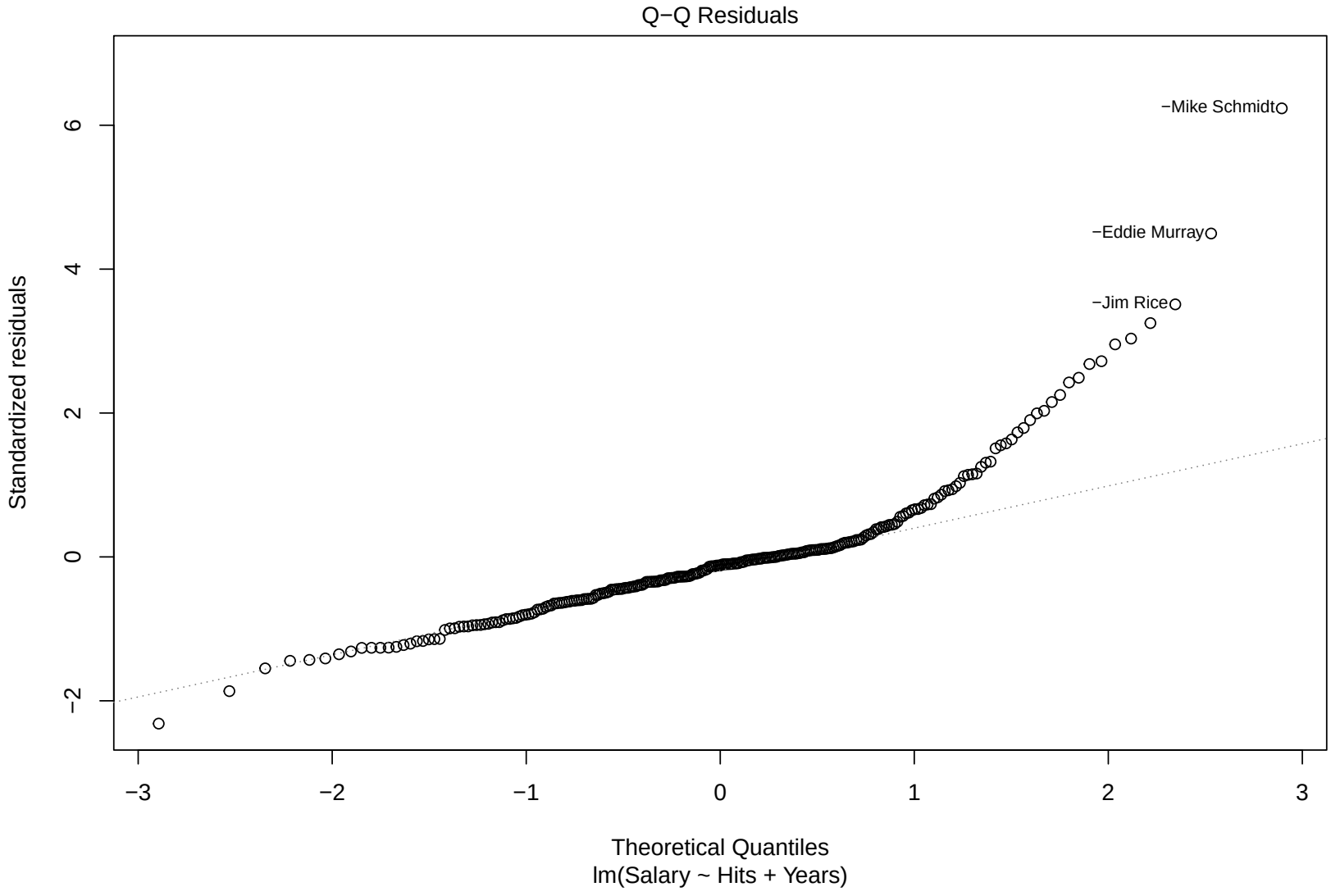
```
        2.5 %   97.5 %
Hits 3.325383 5.299494
```

# Linear Regression Checks

- **Assumption of Normality**: $Y$ remains *normally distributed* across fixed values of $X_1$, $X_2$
- **Assumption of Homoscedasticity**: The variance of $Y$ is *randomly scattered about zero* for different values of $X_1$, $X_2$
- **Limiting Multicollinearity**: $X_1$, $X_2$ should not be correlated with each other, as it diminishes one's ability to interpret a model

# Checking the Normality Assumption

```
plot(lm(Salary ~ Hits + Years, Hitters), 2) # Quantile-Quantile (QQ) Plot
```

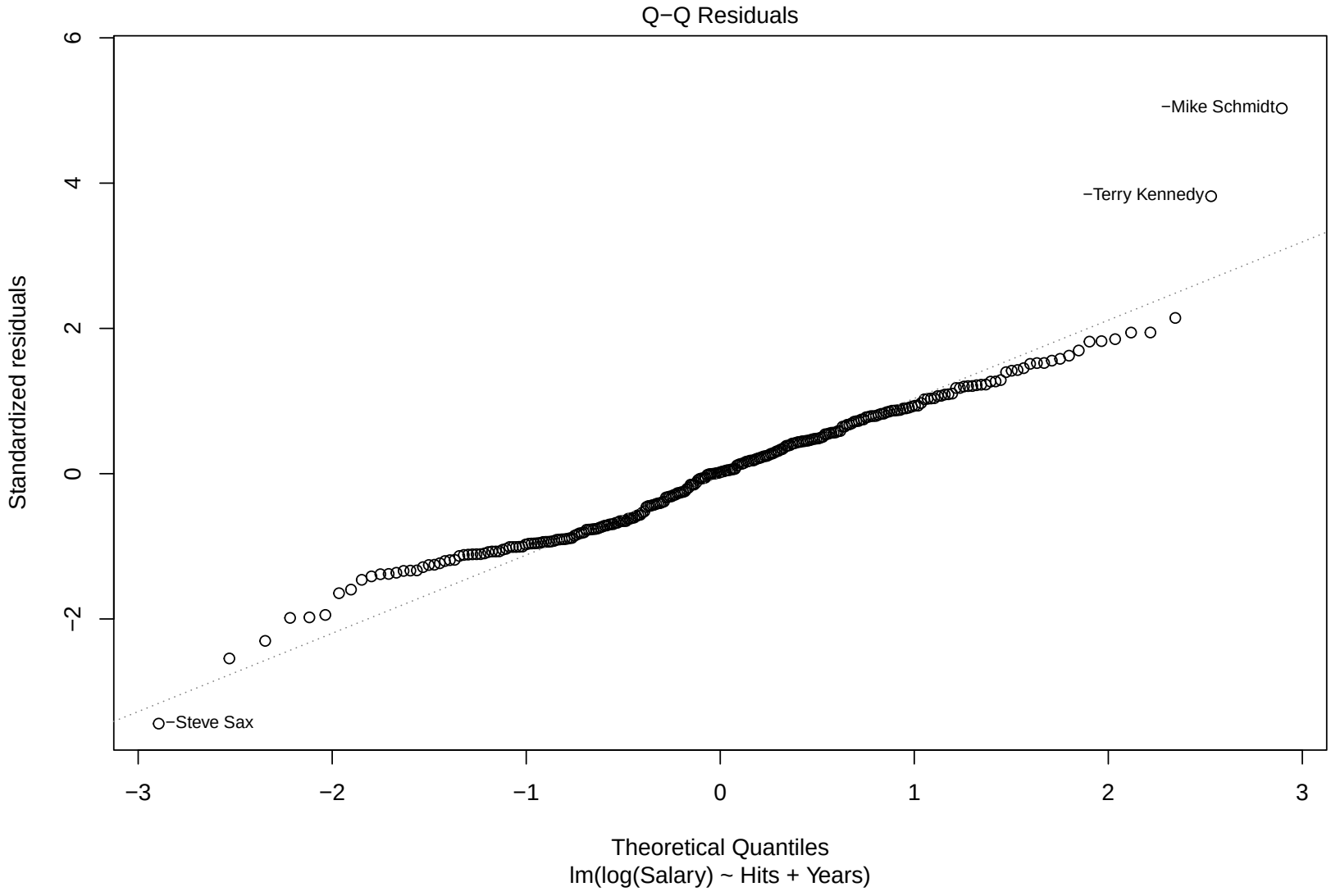


# Checking the Normality Assumption

- The points on the QQ Plot appear to *deviate from* the 45-degree reference line as the theoretical quantiles increase. Therefore, the assumption of normally distributed residuals *is* violated.
- The residuals deviating from the 45-degree reference line as the theoretical quantiles increase indicates a *right-skew*, which can be fixed by applying a *logarithmic* transformation

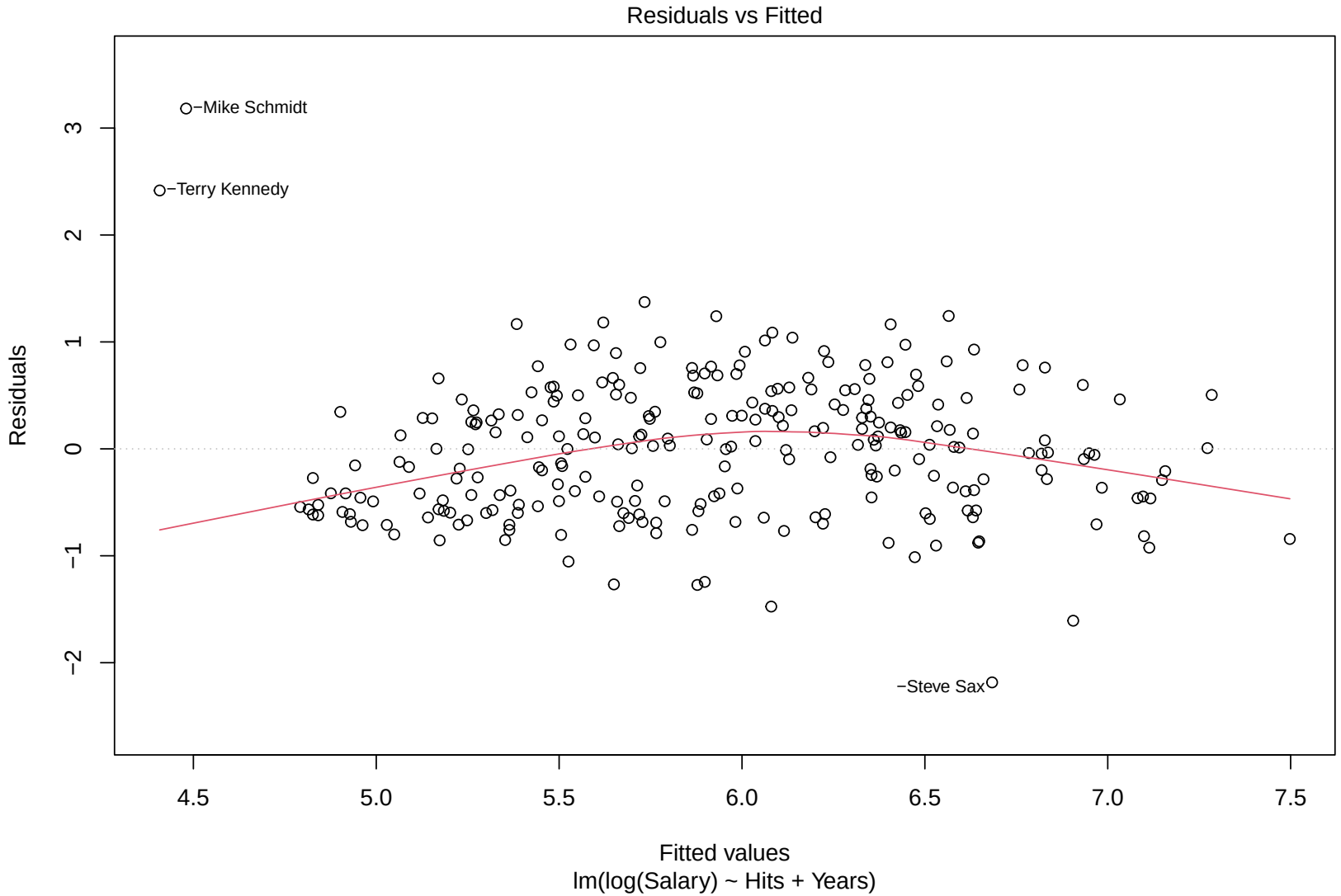
# Fixing Violations in the Normality Assumption

```
plot(lm(log(Salary) ~ Hits + Years, Hitters), 2) # Quantile-Quantile (QQ) Plot
```



# Checking the Homoscedastic Assumption

```
plot(lm(log(Salary) ~ Hits + Years, Hitters), 1) # Residual Plot
```

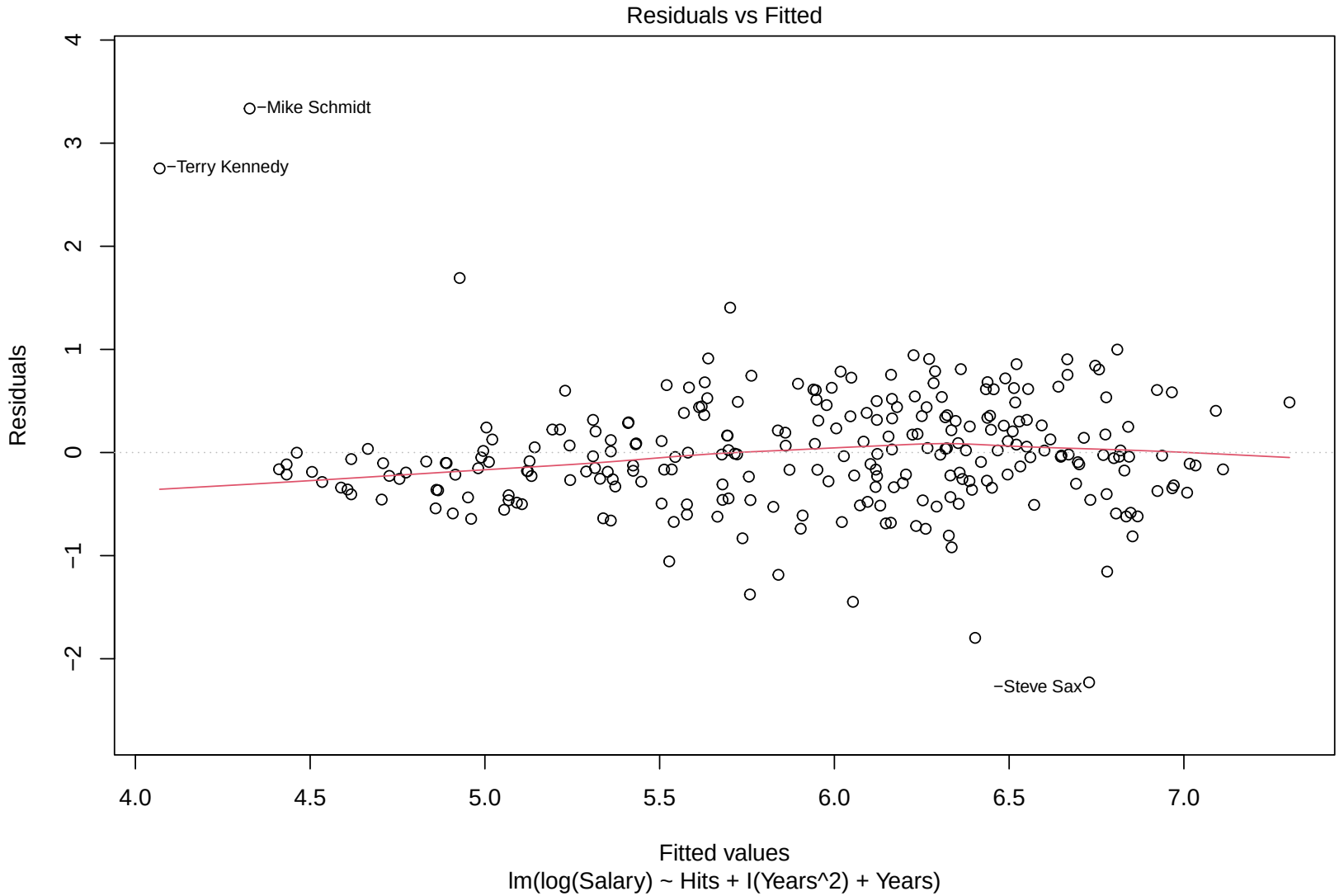


# Checking the Homoscedastic Assumption

- The residuals display a parabolic trend, signaling that the residuals are not *randomly scattered about zero*. Therefore, the assumption of homoscedasticity *is* violated.
- A distinct parabolic trend indicates a *non-linear* relationship within the data, which can be fixed by applying a square transformation to one or more predictors

# Fixing Violations in the Homoscedastic Assumption

```
plot(lm(log(Salary) ~ Hits + I(Years^2) + Years, Hitters), 1) # Residual Plot
```



# Checking for Multicollinearity

- The **variance inflation factor** (VIF) checks multicollinearity
  - Values around 1-5 are good, 5-10 indicates mild multicollinearity, and 10+ is problematic
- In our case, since the squared term is a function of the non-squared term, the VIF appears problematic when it's not

```
library(car)
vif(lm(log(Salary) ~ Hits + I(Years^2) + Years, Hitters))
```

```
      Hits I(Years^2)      Years
  1.046056  13.316620  13.299145
```

```
vif(lm(log(Salary) ~ Hits + Years, Hitters))
```

```
    Hits    Years
1.000346 1.000346
```

# Making Predictions

- You can use an MLR model to evaluate linear relationships and make predictions ($\hat{Y}$) about the response $Y$
- Use the `predict()` function in the `stats` package to store these predictions as a vector

```
mod <- lm(log(Salary) ~ Hits + I(Years^2) + Years, Hitters)
pred <- exp(predict(mod, newdata = Hitters))
sort(pred, decreasing = T)[1:3]
```

```
      -Jim Rice     -Dave Parker -Keith Hernandez
       1483.454         1226.622          1201.644
```

# Other Relevant Information

- **Analysis of Variance (ANOVA)**: Tests whether adding a predictor variable *significantly* improves the model
- **Variable Selection**: Procedures that describe testing for the best combination of predictor variables in model accuracy
  - Stepwise regression adds and/or removes features that lead to improvement in accuracy
  - Least absolute shrinkage and selection operator (LASSO) regression adds a penality that performs variable selection and prevents overfitting